

Trial Evidence Overview

Randomized-trial population and average effect for village-level prioritization

Randomized participants

4,533

Randomized trial population

Observed primary outcome

4,508

99.4% of randomized

Missing primary outcomes

25

0.6% of randomized

Randomized villages

127

64 intervention • 63 control

Adjusted Trial Effect

-1.880

pp

Predicted 10-year ASCVD risk · percentage points

Lower predicted risk favors intervention.

Adjusted Effect and 95% CI

95% CI: -2.564 to -1.196



-2.564 -1.880 -1.196

-3 -2 -1 0 1

Difference in predicted 10-year ASCVD risk (percentage points)

Randomized Allocation Summary

Arm	Participants	Villages
Intervention	2,297	64
Control	2,236	63
Total	4,533	127

Primary follow-up completeness

Status	Participants	Share
Observed	4,508	99.4%
Missing	25	0.6%

Analysis population

- Available-primary-outcome analysis
- Analyzed according to randomized village assignment
- Participants without primary follow-up outcomes are excluded from the endpoint analysis
- Do not describe as unqualified full ITT

Interpretation boundaries

- Endpoint: Predicted 10-year ASCVD risk, not observed cardiovascular events
- Estimate: Average effect of the multicomponent intervention package
- Use: Not an individual clinical recommendation

Risk vs Predicted Benefit

Compare village mean baseline risk with frozen out-of-village predicted intervention benefit signals

Randomized village clusters

127

One mark per randomized village

Baseline risk range

13.07%–23.31%

Village mean predicted 10-year ASCVD risk

GRF benefit range

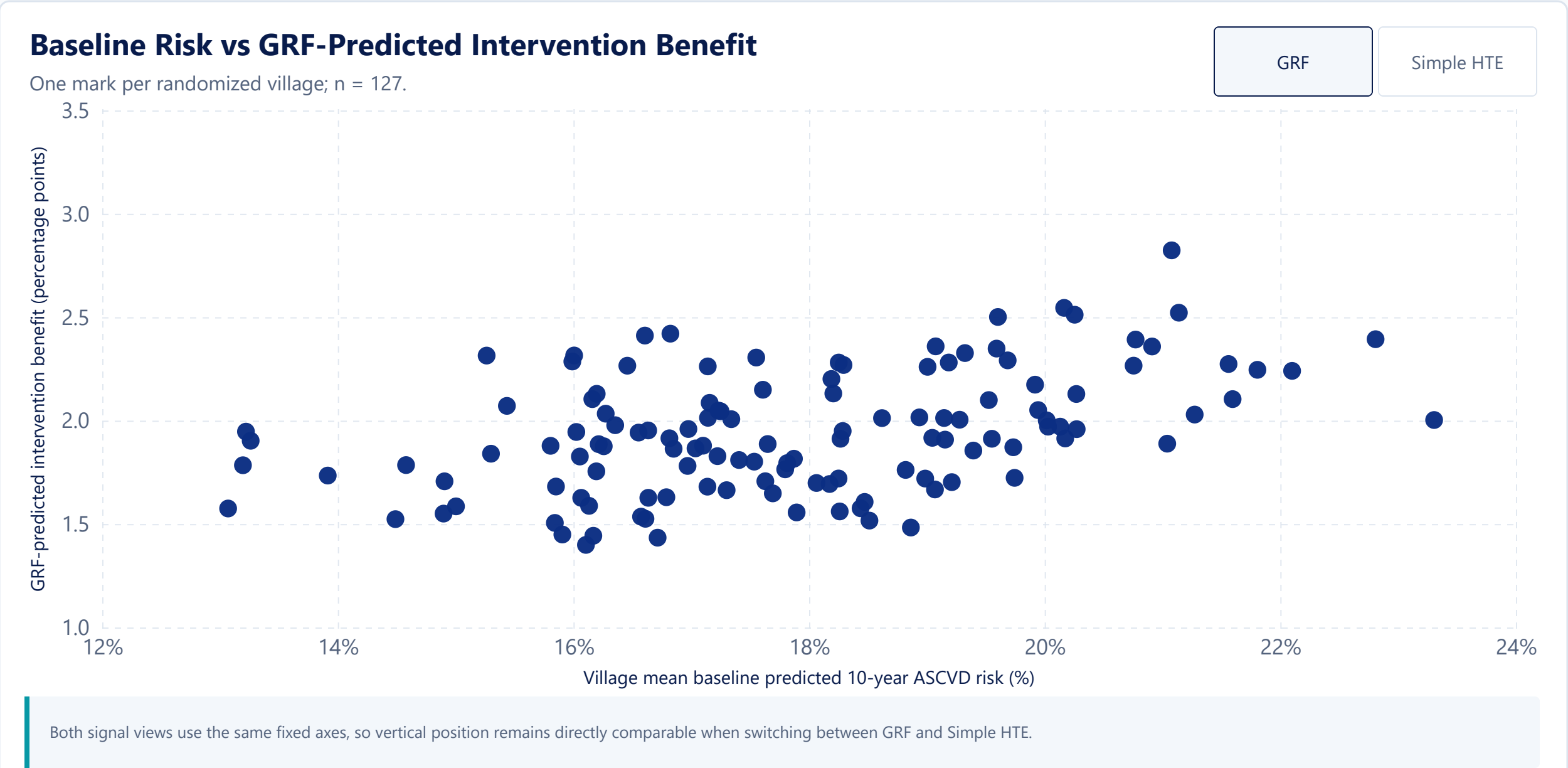
1.399–2.823 pp

Frozen out-of-village predictions

Simple HTE benefit range

1.147–3.277 pp

Frozen out-of-village predictions



Village-level signal agreement

Pairwise Spearman ρ

Comparison	ρ
GRF vs Simple HTE	0.4649
GRF vs Baseline Risk	0.4305
Simple HTE vs Baseline Risk	0.3349

Key insights

- Higher baseline risk does not imply greater predicted intervention benefit.
- GRF and Simple HTE predicted intervention benefit signals are moderately correlated but not identical.
- Baseline risk and predicted intervention benefit are related but not interchangeable targeting signals.

Plot details

- Each mark represents one randomized village cluster.
- Predicted intervention benefits are frozen out-of-village estimates measured in percentage points.
- Positive values indicate a larger predicted risk reduction.
- Correlations are village-level Spearman rank correlations.

HTE Validation

Evaluate prioritization performance, calibration, and classical subgroup evidence

Prioritization validation (AUROC)

Within-fold priorities; point estimates with 95% confidence intervals

GRF	0.6382
Simple HTE	0.7310
Baseline Risk	0.8818
Random	0.0213

● Estimate

— 95% CI

⋮ Reference at 0

Positive values support prioritization above the overall participant population.

Calibration slopes

Fold-adjusted estimates with 95% confidence intervals

GRF	0.916
Simple HTE	0.882

● Estimate

— 95% CI

⋮ Ideal slope = 1

Both 95% confidence intervals include the ideal slope of 1.

Participant-score Ranking Agreement

Pairwise Spearman ρ on raw participant scores

Comparison	ρ
● ● GRF vs Simple HTE	0.6191
● ● GRF vs Baseline Risk	0.7249
● ● Simple HTE vs Baseline Risk	0.6820

Spearman ρ ranges from −1 to 1; agreement does not establish superiority.

Adjusted classical subgroup effects

Available-primary-outcome analysis; lower values favor intervention

Moderator	Subgroup	N	Adjusted effect (95% CI), pp
Overall	Overall	4,508	-1.880 (-2.564 to -1.196)
Age	<60 years	2,688	-1.852 (-2.573 to -1.132)
	≥60 years	1,820	-1.920 (-2.681 to -1.158)
Sex	Women	2,273	-1.613 (-2.343 to -0.882)
	Men	2,235	-2.152 (-2.884 to -1.420)
Education	Below high school	3,556	-1.861 (-2.560 to -1.162)
	High school or above	952	-1.952 (-2.828 to -1.076)
Occupation	Non-agricultural worker	1,462	-2.409 (-3.235 to -1.584)
	Agricultural worker	3,046	-1.629 (-2.344 to -0.913)
Income	≥10,000 yuan	3,792	-1.737 (-2.431 to -1.044)
	<10,000 yuan	716	-2.636 (-3.569 to -1.704)
Baseline risk*	<16%	2,265	-1.186 (-1.917 to -0.456)
		2,243	-2.581 (-3.312 to -1.849)

● Overall estimate

○ Subgroup estimate

— 95% CI

⋮ No-effect reference

Lower favors intervention

Adjusted difference in predicted 10-year ASCVD risk (percentage points)

Evidence summary

GRF prioritization validation

SUPPORTED

Comparative evidence

SUPERIORITY NOT DEMONSTRATED

Interpretation boundaries

AUROC supports prioritization, not effect magnitude or policy value.

Calibration near 1 does not establish superiority.

Subgroup estimates are descriptive; CI overlap is not an interaction test.

Baseline-risk groups are partially reconstructable.

Endpoint: predicted risk, not observed cardiovascular events.

Participant-score agreement is descriptive and does not validate individual treatment selection.

Classical subgroup estimates are descriptive full-data results, not out-of-village prioritization validation.

Rollout Across Capacity

Compare frozen whole-village rollout policies across capacity with pointwise uncertainty.

Explore rollout scenarios

Focused policy

GRF

Simple HTE

Baseline Risk

Capacity anchor (randomized villages)

13

32

64

95

127

Default: Simple HTE · K=64 (starting view, not a recommendation).

Selected villages

64

of 127 randomized villages

Randomized participants covered

2,299

50.7% of 4,533

Population rollout value

1.0784 pp

Pointwise 95% interval: 0.6060 to 1.5518

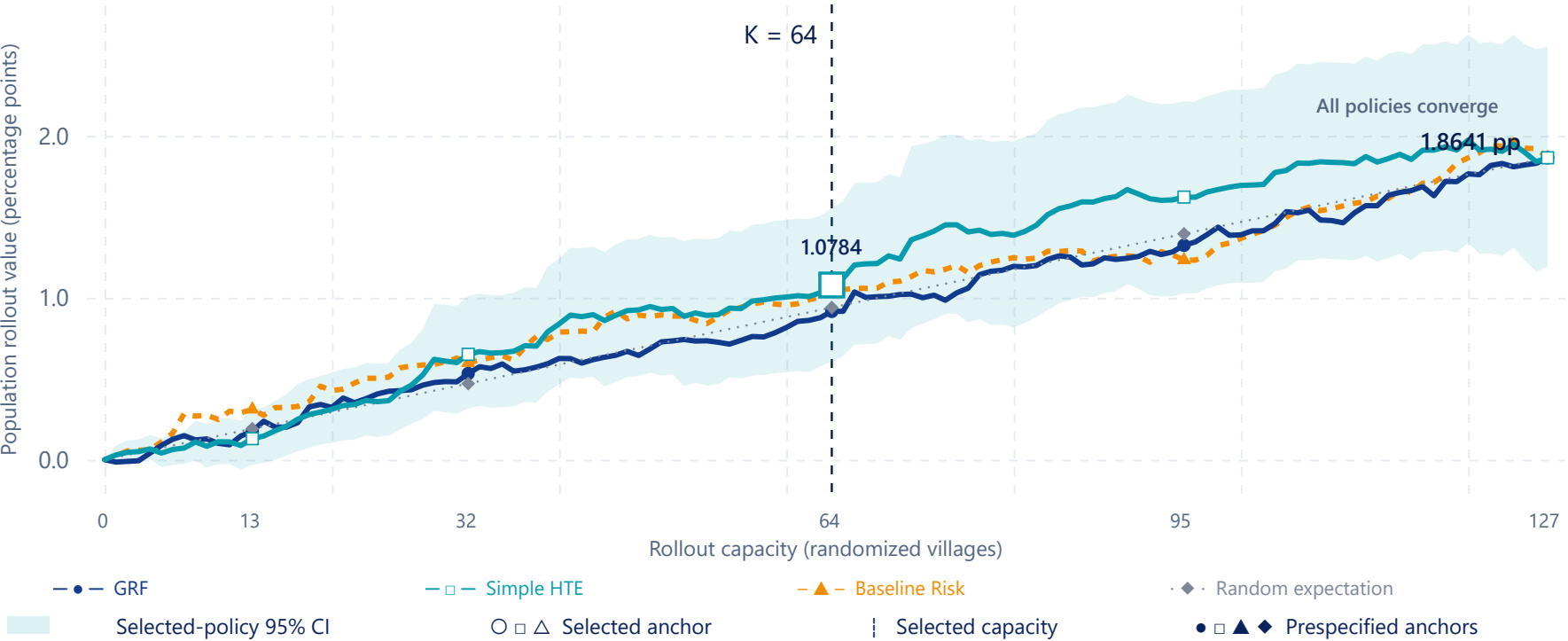
Gain versus random

+0.1390 pp

Pointwise 95% interval: -0.2353 to 0.5103

Population Rollout Value Across Capacity

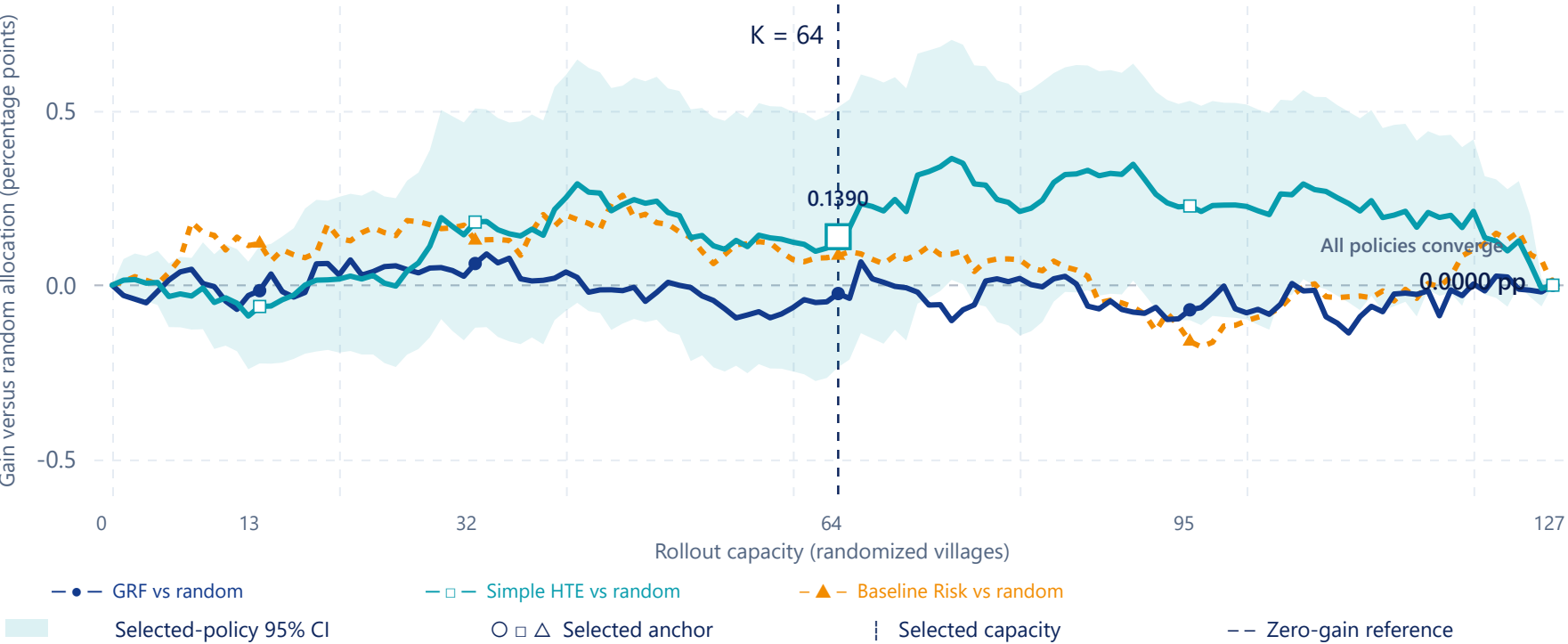
Participant-weighted predicted-risk reduction with pointwise 95% intervals



Pointwise 95% CIs use 2,000 village-bootstrap replicates.

Gain Versus Random Allocation Across Capacity

Policy rollout value minus random-allocation expectation; pointwise 95% intervals



All 12 prespecified gain intervals at K=13, 32, 64, and 95 include zero; K=127 is reconciliation, not an optimum.

Fixed-Capacity Evidence Matrix

Population rollout value (pp) at prespecified anchors

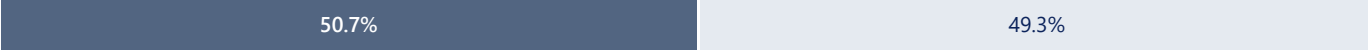
Policy	K = 13	K = 32	K = 64 Selected	K = 95	K = 127 Endpoint
GRF	0.1751	0.5317	0.9153	1.3236	1.8641
Simple HTE	0.1296	0.6507	1.0784	1.6214	1.8641
Baseline Risk	0.3104	0.5980	1.0241	1.2328	1.8641
Random expectation	0.1908	0.4697	0.9394	1.3944	1.8641

Random expectation is reference context, not a selectable rollout policy.

Coverage at Selected Capacity

Simple HTE / 64 randomized villages

Randomized participant coverage



2,299 of 4,533

Not covered: 2,234

Village coverage



64 of 127

Remaining: 63

Observed-outcome evaluation coverage: 2,287 of 4,508 participants (50.7%). Whole randomized villages remain the capacity unit.

Robustness and Village Prioritization

Separate village rankings, policy agreement and sensitivity — no declared winner.

Trial villages

127

One candidate row per village

Lowest agreement (K=64)

42.2%

Simple HTE vs Baseline Risk · 38 shared

Gain intervals including zero

12 of 12

3 policies × 4 prespecified anchors

Ranking-sensitive policies

3

At K=13, 32, 64 and 95

Focused policy

GRF

Simple HTE

Baseline Risk

Capacity anchor (villages)

13

32

64

95

127

Candidate Village Rankings

Click Focus rank for policy order

Three separate policy rankings.

		● GRF		□ Simple HTE		▲ Baseline Risk		
Village ID	Focus rank ▲	Rank	Predicted benefit (pp)	Rank	Predicted benefit (pp)	Rank	Predicted risk (%)	Participants
33	1	24	2.246	1	2.711	2	21.81	35
71	2	106	1.699	2	2.308	56	18.06	35
47	3	2	2.394	3	2.245	3	22.81	35
91	4	63	1.755	4	2.201	110	16.19	35
75	5	47	2.201	5	3.277	52	18.19	38
49	6	7	2.359	6	2.957	31	19.07	39
50	7	37	2.305	7	2.265	66	17.55	33
48	8	83	1.648	8	2.121	69	17.69	38
16	9	10	2.274	9	2.163	10	21.56	38
70	10	35	2.034	10	2.347	100	16.27	38
79	11	16	2.411	11	2.240	96	16.61	32
77	12	72	2.071	12	2.725	107	15.44	40
76	13	33	1.864	13	1.934	83	16.85	40
90	14	29	2.049	14	2.308	74	17.23	37
69	15	40	1.942	15	1.994	103	16.55	36
51	16	17	2.286	16	2.680	97	15.99	31

Policy changes Focus rank and highlight. Click headers to sort whole rows; policy signals stay fixed.

Policy Overlap at K=64

Jaccard agreement · common 0–1 scale

Policy pair	Agreement bar	50% reference	Agreement	Shared
● GRF vs □ Simple HTE			50.6%	43
● GRF vs ▲ Baseline Risk			45.5%	40
□ Simple HTE vs ▲ Baseline Risk			42.2%	38

Robustness at Selected Capacity

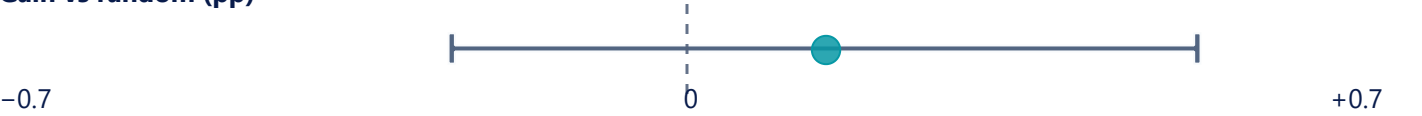
Simple HTE · K=64

+0.1390 pp

Pointwise 95% CI: -0.2353 to 0.5103 pp · includes zero

No clear capacity-specific advantage
Sensitivity checks do not establish an advantage.

Gain vs random (pp)



Check	Finding	Meaning
Raw ranking	Sensitive	Ranking specification
Equal village	Consistent	Equal village weights
Single-village influence	Robust	Leave-one-village-out

Descriptive Coverage by Age

Age

Covered: 2,299 / 4,533

Group	Coverage bar		Coverage	Covered / total	Gap
Under 60 years		Overall 50.7%	51.3%	1,386 / 2,700	+0.6 pp
60 years or older			49.8%	913 / 1,833	-0.9 pp

Descriptive coverage; not a fairness target.